A focus for now, Post-Domain Controller Verification + First AD Administration

Last step we completed: prepared SRV-DC01, worked on static IP/DNS, installed AD DS, planned the new forest/domain, and started DC promotion for kevinlab.local

Since it is promoted, want to answer the question of: Did the DC actually promote correctly, and can I begin managing AD like an admin?

Goals for today:

1. Confirmed the DC is working
2. Checked DNS/domain discovery
3. Opened the main AD management tools
4. Created first OU structure
5. Created first users
6. Created first security groups
7. Started thinking like an AD administrator

Confirm the DC Promotion Worked

Open Windows Server VM and Log In.

Running commands in PowerShell: **hostname, whoami, Get-ADDomain, Get-ADForest, Get-ADDomainController**

Get-ADDomain: shows information about the current Active Directory domain
Get-ADForest: shows information about the entire Active Directory forest
Get-ADDomainController: shows information about domain controllers

Hostname: SRV-DC01
Logged in as: kevinlab/administrator
Domain name: kevinlab
Forest name: kevinlab.local
DC name: SRV-DC01
Any errors:

The server is not just "Windows Server" anymore. It is now ware of the domain, forest, and Domain Controller role.

Check Critical Domain Controller Services

Run: **Get-Service NTDS, DNS,KDC,Netlogon**

NTDS = Active Directory Domain Services
DNS = DNS Server

KDC = Kerberos Key Distribution Center
Netlogon = Domain logon support

```
PS C:\Users\Administrator> Get-Service NTDS,DNS,KDC,Netlogon

Status   Name               DisplayName
------   ----               -----------
Running  DNS                DNS Server
Running  KDC                Kerberos Key Distribution Center
Running  Netlogon           Netlogon
Running  NTDS               Active Directory Domain Services
```

These matter because NTDS is Active Directory itself. DNS helps clients locate the domain and Domain Controller. KDC handles Kerberos tickets. Netlogon supports domain logon and secure channel functions.

DNS and Domain Discovery Check

Need to make sure that DNS is not broken. Run: **ipconfig /all**

DHCP Enabled: No
IPv4 Address: 192.168.1.236
DNS Servers: 127.0.0.1
Primary DNS Suffix: kevinlab.local

Run: **nslookup kevinlab.local**

Nslookup: used to query DNS…so this command asks the DNS server: what IP address does kevinlab.local resolve to?

Nslookup SRV-DC01.kevinlab.local, nltest /dsgetdc:kevinlab.local

Nltest: used to test AD domain connectivity, /dsgetdc:<DomainName> is described as querying DNS for domain controllers and their IP address

Run First DC Health Checks

Run: **dcdiag /test:Advertising,** then **dcdiag /test:DNS**

Dcdiag stands for Domain Controller Diagnostic.
Dcdiag /test:Advertising checks whether the Domain Controller is advertising itself correctly. Is this server telling the domain that it is available as a Domain Controller?
Dcdiag /test:DNS checks whether DNS is working correctly for AD.

Passed Advertising Test. Passed DNS test.

Open Active Directory Tools

In Server Manager: Go to Tools. Open Active Directory Users and Computers.
Useful to identify for later: DNS, Group Policy Management, Active Directory Domains and Trusts, Active Directory Sites and Services

Default containers in kevinlab.local:

- Builtin
- Computers
- Domain Controllers: contains SRV-DC01
- ForeignSecurityPrincipals
- Managed Service Accounts
- Users: contains Administrator

A container and an OU are not the same thing. Default folders like Users and Computers are containers. OUs are more useful for administration because you can link Group Policy to OUs.

Create Your First OU Structure

Right Click Domain Name inside ADUC tools. Click New -> Organizational Unit.

Creating these as top-level OUs:
KEVINLAB-Users
KEVINLAB-Computers
KEVINLAB-Groups
KEVINLAB-Servers
KEVINLAB-Disabled

Inside KEVINLAB-Users, creating:

- IT
- HR
- Finance
- Sales

Inside KEVINLAB-Computers, creating:

- Workstations
- Laptops

Inside KEVINLAB-Groups, creating:

- Security Groups

Create First Test Users

For each user, unchecking user must change password at next logon for ease.

Inside IT, adding a user called Kevin Admin, username: kadmin

In HR, Alice HR, username: ahr

In Finance, Frank Finance, username: ffinance

In Sales, Sam Sales, username: ssales

Create First Security Groups

In Security Groups, Creating:
GG_IT_Admins
GG_HR_Users
GG_Finance_Users
GG_Sales_Users

GG for Global Group.

Using: Group scope: Global, Group type: Security

Adding Users:
kadmin → GG_IT_Admins
ahr → GG_HR_Users
ffinance → GG_Finance_Users
ssales → GG_Sales_Users

Continuing with a second session of the day….

AD Group Model + File Share Permissions is the focus now. I have a bit of a setup here DC -> OUs -> Users -> Global Groups

Next question: How do companies actually give users access to resources?

Users -> Groups -> Permissions -> File shares / resources

Creating a Healthy DC Snapshot

Taking a snapshot before I do anything else. Will be good for a rollback point.

Verify AD Structure With PowerShell

Get-ADDomain

Next: Get-ADOrganizationalUnit -Filter * | Select-Object Name, DistinguishedName
This command lists all OUs in AD and only shows their Name and DistinguishedName.

OUs all visible!

Next: Get-ADUser -Filter * | Select-Object Name, SamAccountName, Enabled

Users all visible!

Next: Get-ADGroup -Filter * | Select-Object Name, GroupScope, GroupCategory

Groups all visible!

Next:
Get-ADGroupMember GG_IT_Admins
Get-ADGroupMember GG_HR_Users
Get-ADGroupMember GG_Finance_Users
Get-ADGroupMember GG_Sales_Users

Hm: Group Membership exposed they weren't in the groups, not sure why it didn't save. Going to try through Powershell instead.

Add-ADGroupMember -Identity "GroupName" -Members "Username"

Total Errors: Group membership, should be fixed now.

Learn the Professional Group Model

AD has group scopes like Global, Domain Local, and Universal. These scopes control what can belong to the group and where the group can be used for access.

Simple Model (AGDLP)
User Account
↓
Global Group
↓
Domain Local Group
↓
Permission

A bad beginner method would be to Give Alice permission directly, then Frank, then Sam. We want to put users into role groups, put role groups into access groups, and assign permissions to access groups.

Create Domain Local Permission Groups

Go to ADUC via Tools again. Navigate to KEVINLAB-Groups then Security Groups. Within creating:
DL_IT_Share_Modify
DL_HR_Share_Modify
DL_Finance_Share_Modify
DL_Sales_Share_Modify

For each: Scope is Domain Local, Group type Security.

Now I'm going to add global groups into the matching domain local groups:
GG_IT_Admins → DL_IT_Share_Modify
GG_HR_Users → DL_HR_Share_Modify
GG_Finance_Users → DL_Finance_Share_Modify
GG_Sales_Users → DL_Sales_Share_Modify

So, GG_ groups describe who the users are. DL_ describe what resource access they receive.

Create Department File Share Folders

New-Item -ItemType Directory -Path "C:\Shares\Departments\IT" -Force
New-Item -ItemType Directory -Path "C:\Shares\Departments\HR" -Force
New-Item -ItemType Directory -Path "C:\Shares\Departments\Finance" -Force
New-Item -ItemType Directory -Path "C:\Shares\Departments\Sales" -Force

Quick confirm: Get-ChildItem "C:\Shares\Departments"

Purpose: Create department resources that will later be accessed by domain users based on group membership.

Apply Basic NTFS Permissions

For each folder, let's apply permissions. Right click on folders -> properties -> security -> advanced.

Add: DL_HR_Share_Modify (specific for HR folder)

Give: Modify, read + execute, list folder contents, read, and write